

LCA 6 – Selecting Impact Category, Function and Functional Unit

ENGEN370

Mark Lay

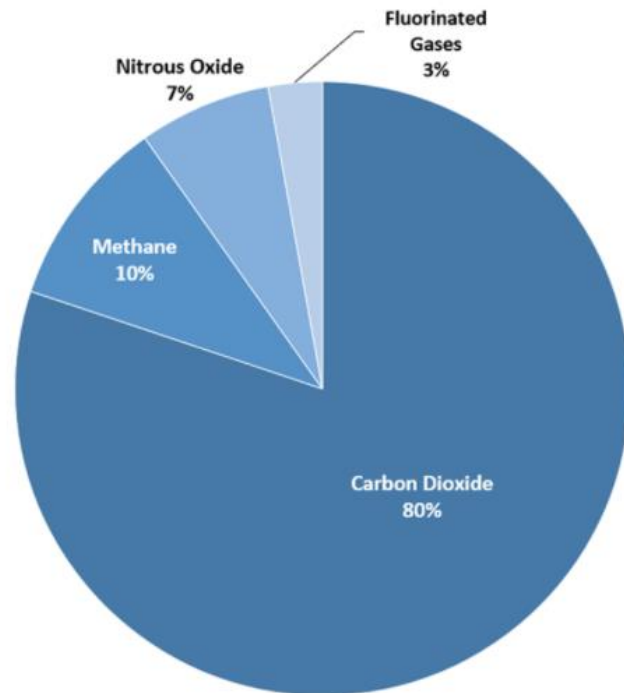
Selecting environmental impacts

<https://ecochain.com/knowledge/impact-categories-lca/>

The ISO Standard requires that a study selects multiple impacts that encompass
“a comprehensive set of environmental issues”

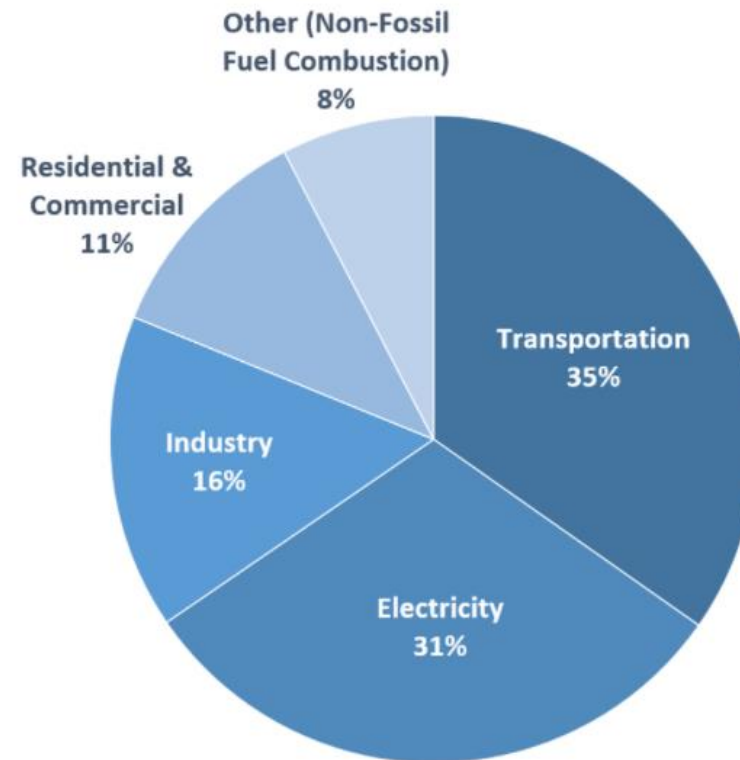
Impact category / Indicator	Unit	Description
Climate change – total, fossil, biogenic and land use (compulsory)	kg CO ₂ -eq	Indicator of potential global warming due to emissions of greenhouse gases to air. Divided into 3 subcategories based on the emission source: (1) fossil resources, (2) bio-based resources and (3) land use change. Calculated by kg of GHG emitted multiplied by a factor to give kg CO ₂ -eq
Ozone depletion	kg CFC-11-eq	Indicator of emissions to air that cause the destruction of the stratospheric ozone layer
Acidification	kg mol H ⁺	Indicator of the potential acidification of soils and water due to the release of gases such as nitrogen oxides and sulphur oxides
Eutrophication – freshwater	kg PO ₄ -eq	Indicator of the enrichment of the fresh water ecosystem with nutritional elements, due to the emission of nitrogen or phosphor containing compounds
Eutrophication – marine	kg N-eq	Indicator of the enrichment of the marine ecosystem with nutritional elements, due to the emission of nitrogen containing compounds.
Eutrophication – terrestrial	mol N-eq	Indicator of the enrichment of the terrestrial ecosystem with nutritional elements, due to the emission of nitrogen containing compounds.
Photochemical ozone formation	kg NMVOC-eq	Indicator of emissions of gases that affect the creation of photochemical ozone in the lower atmosphere (smog) catalysed by sunlight.

Overview of U.S. Greenhouse Gas Emissions in 2019



Total U.S. Emissions in 2019 = 6,558 [Million Metric Tons of CO₂ equivalent](#) (excludes land sector). Percentages may not add up to 100% due to independent rounding.

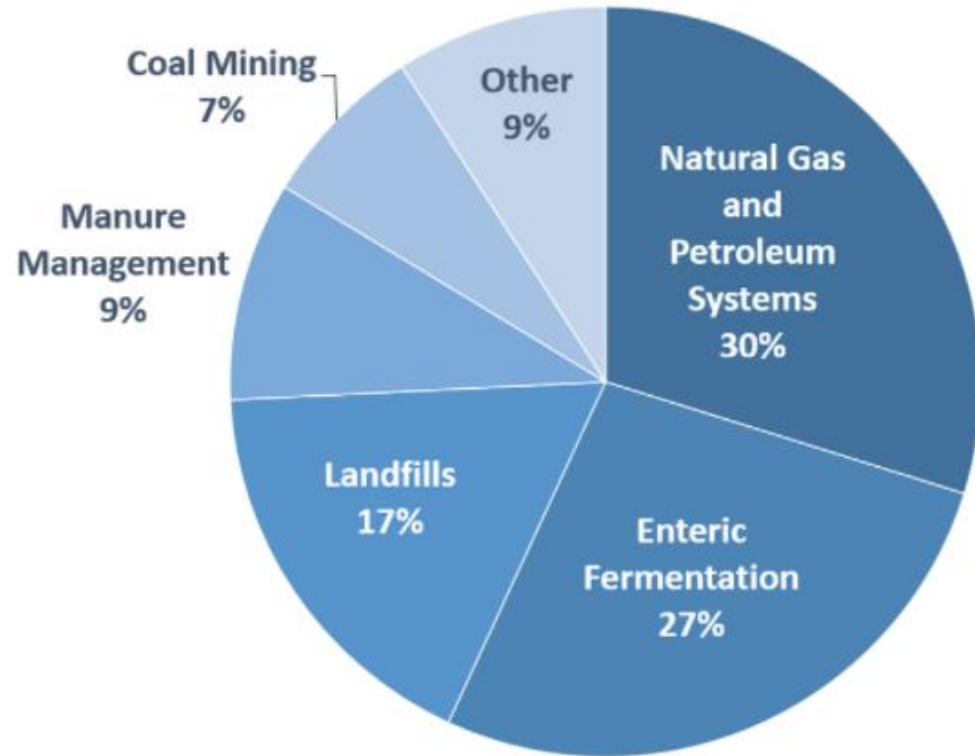
2019 U.S. Carbon Dioxide Emissions, By Source



Note: All emission estimates from the [Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019](#) (excludes land sector).

<https://www.epa.gov/ghgemissions/overview-greenhouse-gases#overview>

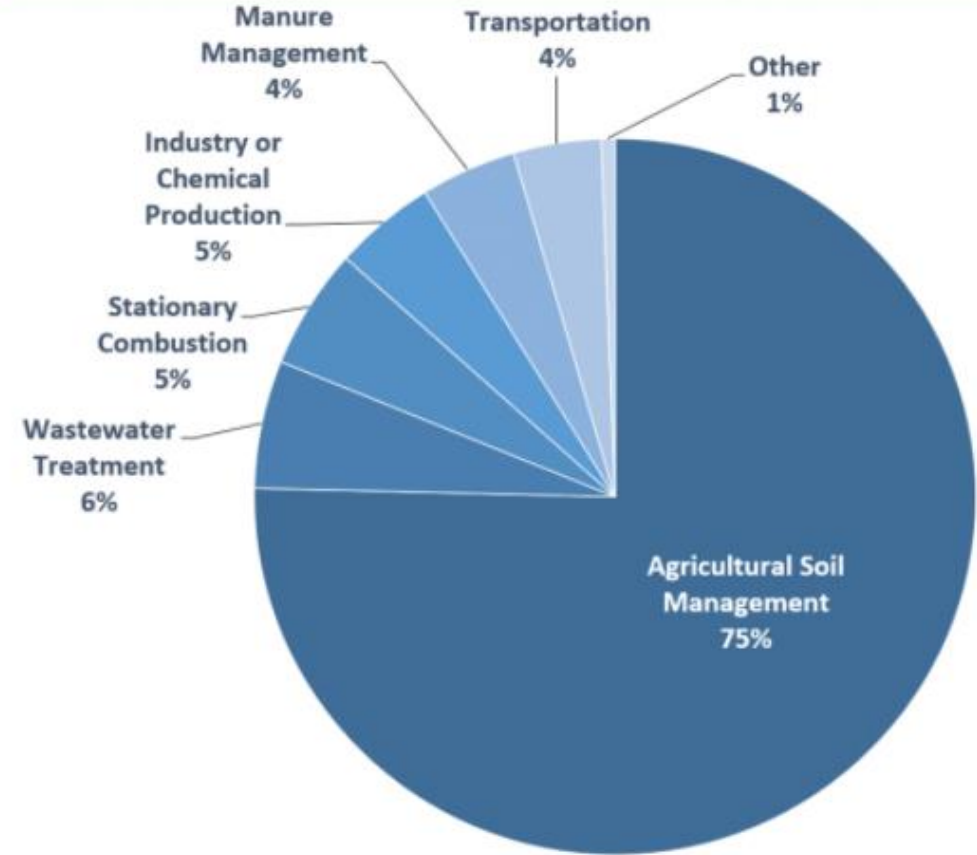
2019 U.S. Methane Emissions, By Source



Note: All emission estimates from the [*Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019*](#) (excludes land sector).

GWP = 25x CO₂

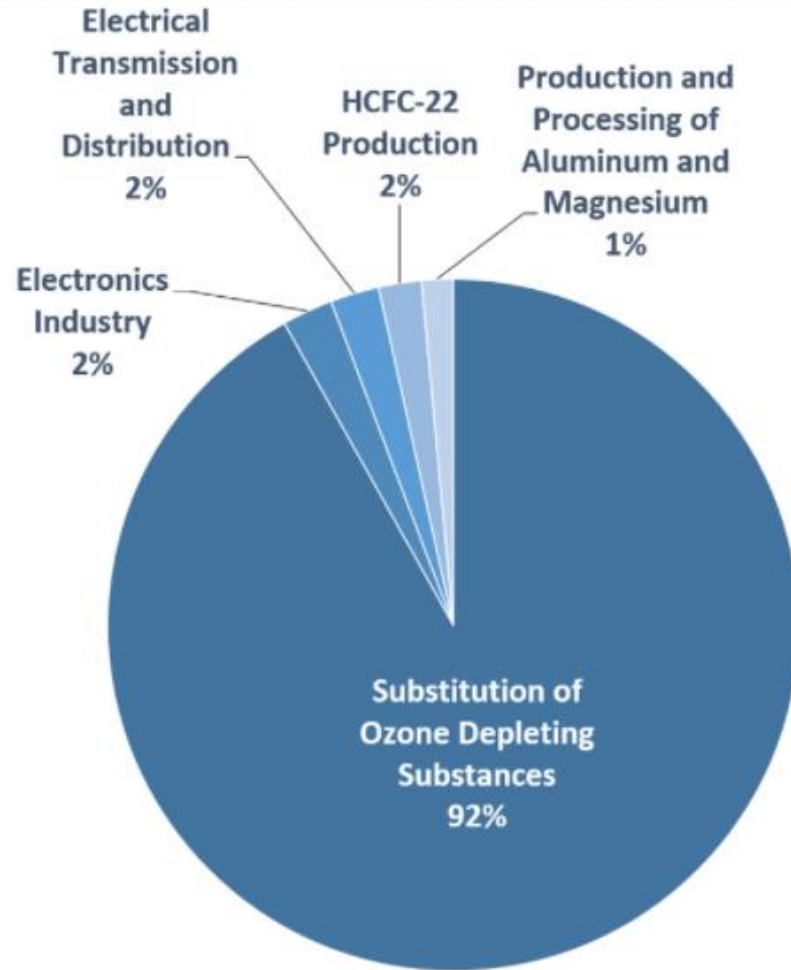
2019 U.S. Nitrous Oxide Emissions, By Source



Note: All emission estimates from the [*Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019*](#) (excludes land sector).

GWP = 298x CO₂

2019 U.S. Fluorinated Gas Emissions, By Source



Note: All emission estimates from the [Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2019](#).

Properties of F-gases

Chemical Formulas:

HFCs, PFCs, NF_3 , SF_6

Lifetime in Atmosphere:

HFCs: up to 270 years

PFCs: 2,600–50,000 years

NF_3 : 740 years

SF_6 : 3,200 years

Global Warming

Potential (100-year):¹

HFCs: up to 14,800

PFCs: up to 12,200

NF_3 : 17,200

SF_6 : 22,800

Selecting environmental impacts

- Data available for calculating GHG emissions (in kg CO₂-eq)
 - <https://www.gov.uk/government/publications/greenhouse-gas-reporting-conversion-factors-2018>
 - Ministry for the Environment. 2020. Measuring Emissions: A Guide for Organisations: 2020 Detailed Guide. Wellington: Ministry for the Environment

Selecting environmental impacts

Impact category / Indicator	Unit	Description
Depletion of abiotic resources – minerals and metals	kg Sb-eq	Indicator of the depletion of natural non-fossil resources. Normalised to Antimony (Sb) by multiplying individual materials by a conversion factor. If only a few abiotic resources you can use kg [material] used.
Depletion of abiotic resources – fossil fuels	MJ, net calorific value	Indicator of the depletion of natural fossil fuel resources. Mass of fossil fuel used (kg) multiplied by the calorific value of the fuel (MJ/kg)
Human toxicity – cancer, non-cancer	CTUh	Impact on humans of toxic substances emitted to the environment. Divided into non-cancer and cancer related toxic substances.
Eco-toxicity (freshwater)	CTUe	Impact on freshwater organisms of toxic substances emitted to the environment.
Water use	m3 world eq. deprived	Indicator of the relative amount of water used, based on regionalized water scarcity factors.
Land use	Dimensionless	Measure of the changes in soil quality (Biotic production, Erosion resistance, Mechanical filtration).
Ionising radiation, human health	kBq U-235	Damage to human health and ecosystems linked to the emissions of radionuclides.
Particulate matter emissions	Disease incidence	Indicator of the potential incidence of disease due to particulate matter emissions.

Selecting environmental impacts

Parameter	Unit	Description
Primary renewable energy (materials)	MJ	Use of renewable primary energy resources as raw materials
Primary renewable energy (energy)	MJ	Use of renewable primary energy, excluding renewable primary energy resources used as energy
Primary renewable energy (total)	MJ	Sum of the two values above
Primary non-renewable energy (materials)	MJ	Use of non-renewable primary energy resources as raw materials
Primary non-renewable energy (energy)	MJ	Use of non-renewable primary energy, excluding renewable primary energy resources used as energy
Primary non-renewable energy (total)	MJ	Sum of the two values above
Use of secondary material	kg	Material recovered from previous use or from waste which substitutes primary materials
Use of fresh water	m3	Freshwater use in absolute values
Use of renewable secondary fuels	MJ	Renewable fuel recovered from previous use or from waste which substitutes primary fuels
Use of non-renewable secondary fuels	MJ	Non-renewable fuel recovered from previous use or from waste which substitutes primary fuels

Deciding on the function of your product/process or infrastructure

- In most cases it is obvious
 - Power plant – produce electricity to supply a country or location
 - Housing – to provide warmth/weather protection for people, store belongings....
 - Watch – to tell you the time
 - Clothing – to provide warmth/weather protection/privacy
 - Vehicles – to transport people/goods from place to place
 - Roading – to allow vehicles to travel from place to place

Deciding on the functional unit

- Depends on the product/process or infrastructure
 - Power plant – e.g. per MJ, kWhr, MWhr, MW
 - Housing – e.g. per house, floor area (m²), or person sheltered
 - Watch – e.g. per item
 - Clothing – e.g. per item (standardised to a particular size)
 - Vehicles – per vehicle, person or kg goods carried per km
 - Roading – per length (e.g. km), capacity (vehicles/day in one direction)
 - Heating – per kW heat output
- Look through other LCAs done in your area to see what they used

Types of LCA

- Attributional – Looking at the impact of one system only (what we are doing)
- Consequential – Comparing effects of decisions made (what we are doing)
 - Choosing between two or three methods of asphalt manufacture
 - Choosing between using recycled concrete or new concrete
- Process LCA – (what we are doing)
- Economic Input Output LCA
 - Aggregates the processes into commodities
 - Environmental impacts of each economic activity
- Hybrid Process LCA, EIO LCA
- Dynamic LCA – includes time dependent, temporal impacts

Wrapping it up

- The goal of your LCA is to investigate and compare the environment impact of [the things you are investigating]. The function of [the things you are investigating] is to [state the function]. The LCA will be a [e.g. gate to gate], attributional, consequential and process LCA, which will include [e.g. the construction and use] of the [thing you are investigating]. The environmental impacts investigated in this study are [state the impacts] and the functional unit used is [state the functional unit]. This LCA is for ENGEN370 – Engineering and the Environment.

Looking back at the project brief

- Provide an overview of your product, infrastructure, or process, what it does, how it works, where it is located.
- Develop a specification table for your product, infrastructure, or process.
- Develop a general flow diagram for the manufacture, use and end of life for your product, infrastructure, or process, including inputs and outputs of raw/processed materials, energy, and wastes (quantities not required).
- Review existing literature to see what information/data is available for your product. infrastructure, or process.
- Define the function (i.e. what it does) of your product, infrastructure, or process.
- Define the functional unit by which you can compare your different products, infrastructures, or processes.

Looking back at the project brief

- Select one or two environmental impacts, one of which must be carbon dioxide (or equivalent) emissions. The other impacts can be non-renewable energy consumption, water consumption, or resource depletion, so choose any that align with your project and goals.
- Choose the scope of your LCA as a group, which may include any of the following:
 - Manufacture (e.g. construction of infrastructure or plant, fabrication of your product, construction of a bridge, building, dam)
 - Use (resources used, emissions and wastes from day to day use)
 - End of use (e.g. demolition, recycling, landfill)